The relationship between landslide susceptibility and landslide affected area represents the landslide hazard. The landslide affected area increases exponentially with susceptibility. For the susceptibility class 0.9-1.0, the landslide rate is ~400 m² year-1 km-2, for the class 0.8-0.9 it is ~150 m² year-1 km-2, and so on.

2.5.3 Results

2.5.3.1 Landslide hazard maps

Looking at the average landslide frequency and landside affected area values (Table 6), we observe the same ranking as for the susceptibility maps. Rwandex-Magerwa is overall the sub-catchment that has the highest average hazard, while Rwabayanga is with the lowest values. However, the comparison of the landslide hazard maps is even less obvious than the susceptibility as it includes the exponential behaviour of the landslide rate component (Figure 6). Therefore, to make the comparison of the hazard maps more robust, we have first calculated the probability density of the susceptibility values (Figure 7 a) that were weighted, using equation of Figure 6, via the multiplication of the landslide rates (Figure 7 b). This gives a better view of the actual hazard in the different sub-catchments, showing the importance of the highest susceptibility classes.

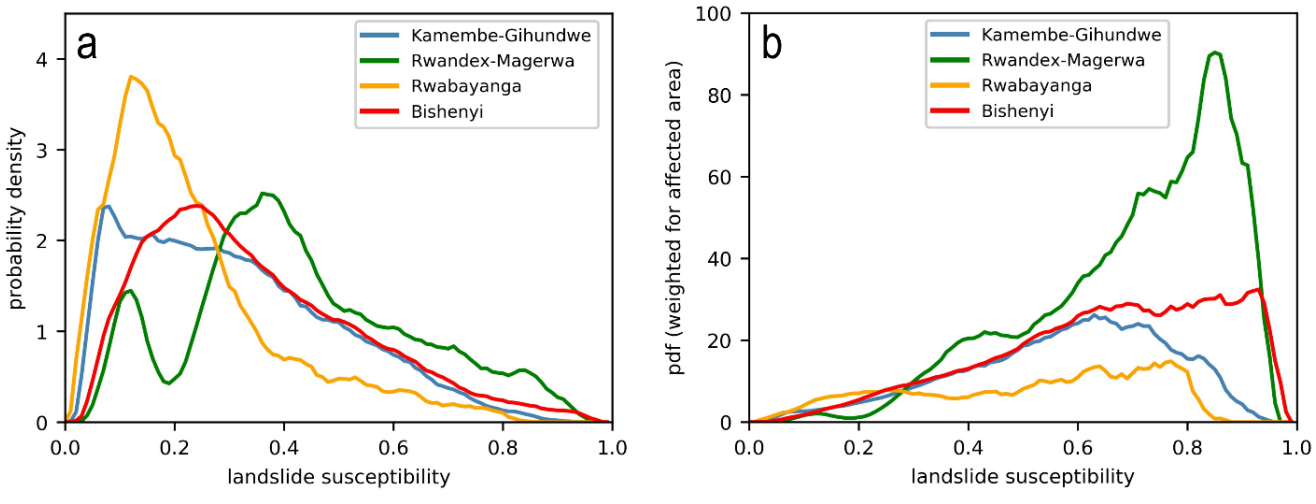


Figure 7: *A) Probability density. B) Probability density weighted for landslide rates (i.e. affected area)*

Therefore, the regional landslide susceptibility model has been classified in four classes, starting from the most susceptible pixel. In other words, the class 0-10% contains the 10% of the landscape that are most prone to landsliding. For each class, the average landslide rate was computed m² year-1 km-2 based on the relationship from Figure 6. The landslide hazard maps are therefore provided for four categories of rates (Figure 8).